

Cough in Children

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Cough helps clear materials from the airways and prevent them from going to the lungs. The materials may be particles that have been inhaled or substances from the lungs and/or airways. Most commonly, material coughed up from the lungs and airways is sputum (also called phlegm—a mixture of mucus, debris, and cells ejected from the lungs).

A cough that brings up sputum is considered productive. Sometimes a cough brings up blood. Some coughs do not bring up anything. They are considered dry or nonproductive. Older children (and adults) typically cough material out, but younger children usually swallow it.

Cough is one of the most common reasons parents bring their children to a health care professional.

(See also [Cough in Adults](#).)

Causes of Cough in Children

Likely causes of cough depend on whether the cough has lasted less than 4 weeks (acute) or more than 4 weeks (chronic).

Common causes

The **most common cause of acute cough** is

- An [upper respiratory infection](#) due to a virus

The **most common causes of chronic cough** are

- [Asthma](#) (the most common)

- [Gastroesophageal reflux](#)
- Postnasal drip (drainage of fluid from the nose down the throat)

Less common causes

Acute cough may also result from a foreign body (such as a piece of food or a piece of a toy) inhaled into the lungs (aspiration) or from less common respiratory infections such as [pneumonia](#), [pertussis](#) (whooping cough), or [tuberculosis](#).

Chronic cough may also result from aspiration of a foreign body, hereditary disorders such as [cystic fibrosis](#) or primary ciliary dyskinesia, a birth defect of the airways or lungs, or [chronic inflammatory disorders](#) involving the airways or lungs. The cough may be stress-related (also known as a habit or psychogenic cough).

AUDIO

[Whooping Cough With the Whoop](#)



Evaluation of Cough in Children

Not every cough requires immediate evaluation by a doctor. Knowing which symptoms may indicate a serious cause can help parents decide whether contacting a doctor is needed.

Warning signs

The following symptoms are of particular concern:

- A blue tint to the lips and/or skin ([cyanosis](#))
- A loud squeaking noise ([stridor](#)) when the child breathes in
- Difficulty breathing
- An ill appearance
- Spasms of uncontrollable, repetitive coughing followed by a high-pitched intake of air (sounds like a whoop)

When to see a doctor

Children who have warning signs should be taken to a doctor or the emergency department right away, as should those whose parents think they may have inhaled a foreign body. If children have no warning signs but have a frequent harsh or barking cough, parents should call the doctor. Doctors typically want to see such children within a day or so, depending on their age, other symptoms (such as fever or sputum), and medical history (particularly a history of lung disorders, such as asthma or cystic fibrosis). Otherwise healthy children who have a cough occasionally and have typical cold symptoms (such as a runny nose) may not need to be seen by a doctor. Contrary

to what many people think, whether sputum is yellow or green or thick or thin does not help a doctor distinguish a bacterial infection from other causes.

Children with a chronic cough and no warning signs should be seen by a doctor, but a delay of a few days to a week is generally not harmful.

What the doctor does

Doctors first ask questions about the child's symptoms and medical history. Doctors then do a physical examination. What they find during the history and physical examination often suggests a cause of the cough and the tests that may need to be done (see table [Some Causes and Features of Cough in Children](#))

Information about the cough helps a doctor determine its cause. Therefore, a doctor may ask

- What time of day does the cough occur?
- What factors—such as cold air, body position, talking, eating, drinking, or exercise—trigger or relieve the cough?
- What does the cough sound like?
- Did symptoms begin suddenly or gradually?
- What are the child's other symptoms?
- Does the cough bring up sputum or blood?

A nighttime cough can be caused by asthma or postnasal drip.

Coughing at the beginning of sleep and in the morning when waking usually is caused by inflammation of the sinuses ([sinusitis](#)). Coughing in the middle of the night is more consistent with [asthma](#).

A barking cough suggests [croup](#) or sometimes a cough that is left over from a viral upper respiratory infection.

A cough that started suddenly in a child with no other symptoms suggests possible inhalation of a foreign body.

Parents of young children and of children with developmental delays are asked about the possibility of swallowing a foreign body (such as a small toy) or small, smooth, firm foods (such as peanuts or grapes). Doctors also ask whether children have had any recent respiratory infections, frequent bouts of pneumonia, allergies, or asthma or has been exposed to tuberculosis or other infections, as may occur during travel to certain countries.

A physical examination is done. To check for breathing problems, doctors observe the child's chest, listen to it with a stethoscope, and tap (percuss) it. Doctors also check for cold symptoms, swollen lymph nodes, and abdominal pain.



Some Causes and Features of Cough in Children

Cause	Common Features*	Tests
Acute cough (lasting less than 4 weeks)		
<u>Bronchiolitis</u>	At first, symptoms of a cold	
	Wheezing and, if bronchiolitis is severe, rapid breathing, with flared nostrils, and difficulty breathing	A doctor's examination
	Possibly vomiting after coughing	Sometimes a chest x-ray and testing of mucus from the nose (taken with a swab) to identify the virus
	Typically in infants up to 24 months old, most often in those 3–6 months old	
<u>COVID-19</u>	Fever, cough, runny nose, rapid breathing, sore throat, shaking chills, headache, loss of taste or smell, vomiting, diarrhea, belly pain, rash, red eyes	Sometimes testing with a nasal swab
	Particularly when there are known cases in the community	Sometimes chest x-ray
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<u>Croup</u>	At first, symptoms of a cold	
	Then a frequent, barky cough (worse at night) and, when croup is severe, a loud squeaking noise when the child breathes in	A doctor's examination Sometimes a neck and chest x-ray